

Data Analysis

Interpret teacher-provided solution requirements and designs, analyse data, and develop data visualisations to present findings.

The big picture

This outcome follows one dataset all the way from collection to a finished visualisation. You decide what data you need and whether it is qualitative or quantitative, judge its quality, handle it lawfully under the Australian Privacy Principles, structure it in a database or spreadsheet, manipulate and validate it, summarise it with descriptive statistics, and finally communicate your findings through a clear, well-designed visualisation. Every key knowledge dot point is a stage in that pipeline — so understanding how the stages connect matters as much as knowing each one.

How it's assessed

School-assessed for Units 1&2, but the skills are examined the same way VCAA examines them in Unit 3&4: precise definitions, correct use of terminology, and applying a concept to the specific scenario in front of you. Expect to be asked to classify data, name and apply a specific Australian Privacy Principle, design a database/spreadsheet structure, choose and justify a validation technique, calculate or interpret a descriptive statistic, and evaluate a visualisation against its purpose and audience. Marks are lost by being vague, naming the wrong principle, or answering in general terms instead of about the scenario.

Key knowledge at a glance

KK	Topic	In one line
KK01	Qualitative & quantitative data	Before you can analyse anything, you have to know what kind of data you're holding — and why you bothered to collect it. This is the first decision in every data project, and examiners love it because it's the one students rush.
KK02	Characteristics of data types	Every value you store has a type — and the type you choose decides what you're allowed to do with it. Pick the wrong one and your phone numbers lose their leading zeros, your dates become nonsense, and your totals quietly break.
KK03	The quality of data & information	A data visualisation is only ever as trustworthy as the data underneath it. This module unpacks the five factors VCAA uses to judge that trust — accuracy, bias, integrity, relevance and reliability — and the sub-factors hiding inside one of them.

KK	Topic	In one line
KK04	Characteristics of data & information	Six properties decide whether a dataset is actually fit for the job : size, structure, relevance, accessibility, clarity and context. Get them right and raw data becomes useful information. Get them wrong and even "correct" data is useless.
KK05	Australian Privacy Principles	When someone hands you their personal data, you don't own it — you hold it in trust. The Privacy Act 1988 (Cwlth) sets out 13 Australian Privacy Principles (APPs) that say how organisations must handle that trust. Your study design zooms in on three of them.
KK06	Ethical issues around data & information	Data isn't just numbers in a spreadsheet — it's people. This dotpoint is about the line between what you can do with data and what you should do.
KK07	Referencing primary & secondary data with APA	When you use someone else's data, words or ideas in a data-analysis project, you have to say where they came from. The APA system is the agreed way to do that — so your reader can trust your work and trace every source.
KK08	Requirements, constraints & scope	Before you build a single chart, you have to answer four questions: what must it do, how well must it do it, what limits are you working inside, and where does the solution stop ? Get these right and the whole data-analysis project has a spine.
KK09	Design tools for solutions	Imagine being told to build a house. You wouldn't start pouring concrete and hope it works out. You'd get a blueprint first — a plan everyone can read, check and agree on before a single brick is laid.
KK10	KK10 · Relational Database Structure — Digital Vector	When the student council started selling hoodies online, they kept everything in one giant spreadsheet. By week three it was a disaster — Maya's email was spelled three different ways and nobody could update a price without breaking something. This module shows how splitting that chaos into tables joined by keys fixes
KK11	Spreadsheet structure: columns, rows and cells	Every chart, total and dashboard you'll ever build in this course starts life as a grid. Before you can analyse data, you have to know exactly how a spreadsheet is built — because the structure is what makes data findable, sortable and safe to calculate with. This dot point is about that architecture: the parts of the
KK12	Manipulating & validating data	Raw data is almost never ready to use. Before it can tell a story, you have to manipulate it into shape, validate that it's clean, and test that everything you built actually works. These are the engine-room skills of data analysis.
KK13	Descriptive statistics	Turning a pile of raw numbers into a few honest summary figures — using spreadsheet functions to calculate the average, median, count/frequency and standard deviation, and knowing which one actually answers the question.

KK	Topic	In one line
KK14	Why are you making this chart?	Before a single colour is chosen or a single bar is drawn, every good data visualisation answers one question: what is it for? This module is the 14th key-knowledge point of the Data Analysis chapter — the purposes of data visualisations.
KK15	Three ways to show the same story	You've cleaned the data, run the stats, found the pattern. Now comes the decision that earns marks: which type of data visualisation do you actually build? The same numbers can become a tall scrolling infographic, a wall of posters, or a live dashboard — and picking the wrong one for the audience loses the message.
KK16	The three things every visualisation is built from	Open any infographic, dashboard or poster and pull it apart. Underneath, it is always made of the same three component families: text & graphics, tables, and charts & graphs. Knowing what each one is for — and when it is the wrong choice — is what the exam rewards.
KK17	Formats & conventions	The same data can look professional and instantly readable — or messy and confusing. The difference is rarely the numbers. It's the rules you apply consistently: how you name things behind the scenes, and how you use colour, fonts, images and icons out front.

Chapter glossary

Term	Meaning
Qualitative data	Data describing qualities or categories (e.g. opinions, colours) that cannot be meaningfully averaged.
Quantitative data	Numeric data that can be measured and calculated with (e.g. counts, times, scores).
Data integrity	The accuracy, completeness and consistency of data over its whole life cycle.
Australian Privacy Principles (APPs)	The 13 principles in the Privacy Act 1988 (Cwlth) governing how personal information is handled.
Primary key	A field (or combination) that uniquely identifies each record in a table.
Foreign key	A field in one table that refers to the primary key of another, forming a relationship.
Validation	Checking that entered data is reasonable and complete (existence, type, range).
Standard deviation	A measure of how spread out values are around the mean.
Data visualisation	A graphical representation of data designed to inform, educate, persuade or entertain an audience.

Answering across the chapter

These scenario questions pull together several key-knowledge points at once — the kind of integrated thinking the exam rewards. Compare the weak and strong responses.

Q1. Apply · 4 marks

A school surveys students about canteen choices and stores their names, year levels and answers in a spreadsheet to build a public dashboard. Identify two data-handling issues across this pipeline and explain how each should be addressed.

✗ A weak answer

They should keep the data private and make the dashboard look nice. (→ *Names no principle, no validation, no structure — generic, 0-1.*)

✓ A strong answer

First, **privacy**: the dashboard is public, so under APP 6 (use/disclosure) and APP 2 (anonymity) the names should be removed and the data aggregated, since student identity is not needed to show canteen trends. Second, **data quality/validation**: year level should be validated (e.g. range 7-12) so invalid entries don't distort the counts. Addressing both keeps the data lawful and the findings accurate.

How the marks are awarded

- 1 mark — identifies a privacy issue and names a relevant APP.
- 1 mark — gives an appropriate fix (de-identify/aggregate).
- 1 mark — identifies a data-quality/validation issue.
- 1 mark — gives an appropriate validation fix.

Q2. Justify · 3 marks

Justify the choice of a relational database rather than a single spreadsheet for storing student, class and result data.

✗ A weak answer

Databases are better than spreadsheets because they hold more data. (→ *Capacity isn't the real reason; ignores redundancy/relationships — 1.*)

✓ A strong answer

A relational database stores students, classes and results in separate tables linked by **primary and foreign keys**, so each fact is stored once. This removes the duplicated data a single spreadsheet would carry, reduces update errors, and lets you run queries across related tables. The structure protects **data integrity** in a way a flat spreadsheet cannot.

How the marks are awarded

- 1 mark — separate tables linked by keys.
- 1 mark — reduced redundancy / single source of truth.
- 1 mark — links to integrity or querying benefit.

Chapter revision checklist

- Classify data as qualitative or quantitative and name its type (text, numeric, Boolean).
- Judge data quality using accuracy, bias, integrity, relevance and reliability.
- Name and apply the correct Australian Privacy Principle to a scenario.
- Acknowledge sources using the APA referencing system.
- Design a relational database (tables, primary/foreign keys, relationships) and a spreadsheet structure.
- Choose and justify validation techniques and manipulation functions for a dataset.
- Calculate and interpret average, median, count/frequency and standard deviation.
- Select a visualisation type for a given purpose and audience and evaluate its components.